

ScoreActuary: Hoop-Centric Trajectory-Aware Network for Fine-Grained Basketball Shot Analysis

Ting-Yang Kao
Inst. of Info. Sys. and Appl., National
Tsing Hua University, Taiwan
gs109065516@gapp.nthu.edu.com.tw

Tse-Yu Pan
Dept. of Computer Science, National
Tsing Hua University, Taiwan
typan@mx.nthu.edu.tw

Chen-Ni Chen
Dept. of Computer Science, National
Tsing Hua University, Taiwan
cn.chen@mx.nthu.edu.tw

Tsung-Hsun Tsai
Dept. of Computer Science, National
Tsing Hua University, Taiwan
109062566@gapp.nthu.edu.com.tw

Hung-Kuo Chu
Dept. of Computer Science, National
Tsing Hua University, Taiwan
hkchu@cs.nthu.edu.tw

Min-Chun Hu
Dept. of Computer Science, National
Tsing Hua University, Taiwan
anitahu@cs.nthu.edu.tw

ABSTRACT

We propose a fine-grained basketball shot analysis system called ScoreActuary to analyze the players' shot events, which can be applied to game analysis, player training, and highlight generation. Given a basketball video as input, our system first detects/segments shot candidates and then analyzes "Shot Type", "Shot Result", and "Ball Status" of each shot candidate in real-time. Our approach is composed of a customized object detector and a trajectory-aware network to learn the information of ball trajectory. Compared to the existing methods that analyze basketball shots, our algorithm can better handle videos with arbitrary camera movements while improving the accuracy. To the best of our knowledge, this work is the first system that can analyze fine-grained shot events accurately in real basketball games with arbitrary camera movements.

CCS CONCEPTS

• Computing methodologies → Computer vision.

KEYWORDS

ball detection, hoop detection, video analysis, shot analysis

ACM Reference Format:

Ting-Yang Kao, Tse-Yu Pan, Chen-Ni Chen, Tsung-Hsun Tsai, Hung-Kuo Chu, and Min-Chun Hu. 2022. ScoreActuary: Hoop-Centric Trajectory-Aware Network for Fine-Grained Basketball Shot Analysis. In *Proceedings of the 30th ACM International Conference on Multimedia (MM '22)*, October 10–14, 2022, Lisboa, Portugal. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3503161.3547736>

1 INTRODUCTION

In recent years, multimedia technologies have been widely employed for game analysis and player training in the field of sports. Among the existing applications that take advantage of video analysis, HomeCourt [3] is the most well-known interactive app for personal basketball training, offering functionalities such as shot

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
MM '22, October 10–14, 2022, Lisboa, Portugal
© 2022 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9203-7/22/10.
<https://doi.org/10.1145/3503161.3547736>

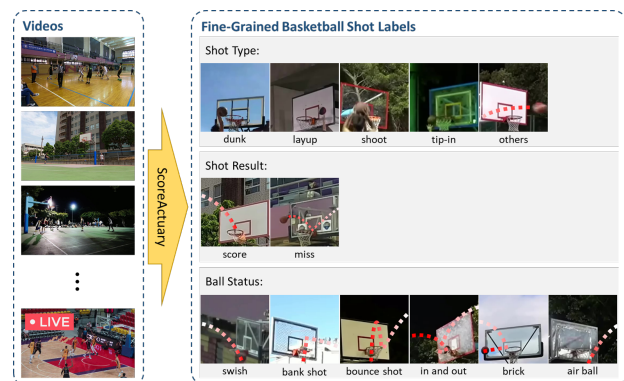


Figure 1: Input and label examples of ScoreActuary.

analysis (e.g., shooter localization, ball tracking, and score detection) and dribble workout.

In addition to HomeCourt, there exists other video-based shot analysis methods [1, 2, 5]. However, all the existing systems/methods [1–3, 5] have the following problems. (1) When the ball trajectory in the 2D camera view passes the area around the hoop in a non-shooting event such as passes, the systems tend to wrongly consider it as a shot event. (2) For a missing shot whose 2D trajectory overlaps the hoop, most methods/systems usually wrongly classify it as a score rather than a miss. (3) None of the existing shot analysis methods/systems can be applied to analyze videos captured with large camera movements (e.g., real-time broadcasting basketball videos). Most methods [1, 2, 5] can only be applied to videos captured by stationary cameras. HomeCourt allows certain camera movements, but cannot handle the case of zooming well.

In this work, we propose a fine-grained basketball shot analysis system, ScoreActuary, which is able to analyze videos recorded using arbitrary viewing angles and camera movements. By focusing on the area around the hoop to acquire the ball trajectory of a shot candidate, our system can deal with the above-mentioned issues to analyze the shot candidates with fine-grained labels. Figure 1 illustrates some examples of the input videos and all the fine-grained labels our system can detect. Each shot candidate has one or three fine-grained labels: (1) shot type (i.e., dunk, layup, shoot, tip-in or others), (2) shot result (i.e., score or miss), and (3) ball status (i.e., swish, bank shot, bounce shot, in and out, brick, or air

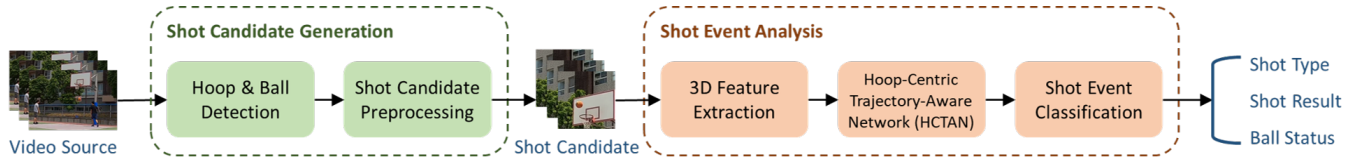


Figure 2: Proposed system framework.

ball). Note that if the shot type of the shot candidate is considered as "others", there would be no shot result and ball status labels due to it is not a shot event. The fine-grained labels can not only extend the flexibility for automatic highlight generation but also improve the awareness of players' training status by providing more detailed shot information. To sum up, the main contribution of our work is that we propose a real-time basketball shot analysis system which can provide fine-grained shot labels with the state-of-the-art performance, especially for basketball videos captured by arbitrary viewing angles and camera motions.

2 SYSTEM FRAMEWORK

Our proposed framework is shown in Figure 2. We observe that the results of shot event analysis highly depend on the ball movements around the hoop. Thus, we first detect where the ball and the hoop are to form a shot candidate, and then model the ball trajectory in the shot candidate to further analyze the shot event.

For shot candidate generation, YOLOR [6] is utilized to construct a hoop detector and a basketball detector. We collected 2,054 images of basketball games and manually annotated hoop/basketball regions in them to train the YOLOR detectors. Afterwards, to avoid the interference of players while extracting features of the ball trajectory, we only use a cropped area near the hoop (instead of the overall frame) to detect the ball and generate a clip for shot analysis. Since the shot candidate is formed by a sequence of near-hoop areas aligned based on the hoop positions, our method is able to analyze videos with various camera movements. However, sometimes the hoop may not be successfully detected when it is occluded by players or referees standing very close to the camera. In this case, we apply linear interpolation to the hoop positions along time to fill up the missing frames.

For each generated shot candidate, the shot event analysis component is applied to classify fine-grained events such as shot type, shot result, and ball status, offering more diverse choices for automatic highlight generation. On the other hand, with the aid of shot type and ball status, players can figure out their training status more accurately under the same number of scored/missed shots. The shot event analysis component aims to model not only the ball trajectory but also the relationship between the ball and the hoop by extracting the spatio-temporal information of the shot candidate. Therefore, we propose a hoop-centric trajectory-aware network (HCTAN), which is inspired by the relation network in Actor-Centric Relation Network (ACRN) [4]. ACRN learns the relationship between actors and background features in order to deal with the action detection task. In our work, we consider the hoop as an actor and utilize the concept of relation network to learn the ball trajectory information with respect to the hoop.

Table 1: Accuracy (%) comparison of shot events. ("SD", "ST", "SR" and "BS" represent "Shot Detection", "Shot Type", "Shot Result", and "Ball Status", respectively.)

Method	SD	ST	SR	BS
ZEPP Standz Basketball [2]	88.54	-	88.81	-
HomeCourt [3]	90.73	-	95.59	-
ScoreActuary	97.35	94.12	100.0	88.24

3 RESULTS

System Evaluation. To evaluate our system, we first collected a basketball shot dataset containing videos from more than 50 different courts and arbitrary recording angles. There are 6,137 shot candidates used to train our proposed HCTAN. Afterwards, we compared the accuracy of shot detection and the three shot event categories against the two existing applications, ZEPP Standz Basketball [2] and HomeCourt [3]. We collected five videos for evaluation, and the videos are recorded from three real 5 on 5 basketball games which include 151 shot candidates. Table 1 indicates that our system outperforms the other two applications in terms of Shot Detection and Shot Result. Our method also has superior accuracy in identifying fine-grained events such as Shot Type and Ball Status, which cannot be achieved by ZEPP Standz Basketball and HomeCourt. In real basketball games, the hoop is often occluded by players or referees. Furthermore, the ball often flies through the hoop during passes. With the enhancement of our system, we are confident that ScoreActuary is the first system that can analyze actual shot events in real basketball games.

Execution time. We deployed ScoreActuary onto a server equipped with a Nvidia GeForce RTX 3090 GPU. It took less than 0.05 seconds to infer a shot candidate, and the overall FPS surpassed 130 after applying TensorRT to YOLOR detectors, making our system be capable of running in real-time.

4 CONCLUSION

In this paper, we introduce a basketball shot analysis system, ScoreActuary, to detect fine-grained shot events in real-time with the state-of-the-art performance. The proposed ScoreActuary is capable of dealing with basketball videos captured by arbitrary viewing angles and camera movements, and therefore it can be used to analyze broadcasting basketball game videos. In the future, we will focus on game data analysis for player training and highlight generation. More details about ScoreActuary can be viewed at <https://youtu.be/tAY4gIYRazk>.

REFERENCES

- [1] Xu-Bo Fu, Shao-Long Yue, and De-Yun Pan. 2021. Camera-based basketball scoring detection using convolutional neural network. *International Journal of Automation and Computing* 18, 2 (2021), 266–276.
- [2] Wu Liu, Chenggang Clarence Yan, Jiangyu Liu, and Huadong Ma. 2017. Deep learning based basketball video analysis for intelligent arena application. *Multimedia Tools and Applications* 76, 23 (2017), 24983–25001.
- [3] Nex Team Inc. 2017. *HomeCourt*. Retrieved May 11, 2022 from <https://www.homecourt.ai/>
- [4] Chen Sun, Abhinav Shrivastava, Carl Vondrick, Kevin Murphy, Rahul Sukthankar, and Cordelia Schmid. 2018. Actor-centric relation network. In *Proceedings of the European Conference on Computer Vision (ECCV)*. 318–334.
- [5] Osman Murat Teket and Imam Samil Yetik. 2020. A Fast Deep Learning Based Approach for Basketball Video Analysis. In *Proceedings of the 2020 4th International Conference on Vision, Image and Signal Processing*. 1–6.
- [6] Chien-Yao Wang, I-Hau Yeh, and Hong-Yuan Mark Liao. 2021. You only learn one representation: Unified network for multiple tasks. *arXiv preprint arXiv:2105.04206* (2021).